E-commerce Sales Data Analytics

Final Data Analytics Capstone — Project Report Template

Dataset integrity note: This starter report includes demo calculations from a simulated practice dataset. Replace all demo figures, findings, and charts with results from the public Sample Superstore dataset before submission.

1. Project Overview

This project demonstrates an end-to-end analytics workflow: dataset selection, data quality checks, cleaning, exploratory analysis, visualization, Power BI reporting, and business recommendations.

2. Problem Statement

Retail stakeholders need a consistent view of sales, profitability, product mix, customer segments, regional performance, and monthly trends to support data-informed decisions.

3. Dataset Description

Recommended source: public educational Sample Superstore dataset. Expected fields include order dates, ship dates, region, category, sub-category, segment, sales, quantity, discount, and profit. The included CSV is a simulated demo dataset used only to make the starter code reproducible.

4. Data Preparation

Inspect data types, missing values, duplicates, numeric validity, and date fields. Remove exact duplicates; parse dates; validate sales and quantity; impute missing categorical values using the mode, city as Unknown, and discount using the median. Record any additional changes made to the real dataset.

5. Exploratory Data Analysis

Calculate descriptive statistics, sales and profit by category/region/segment, monthly sales trends, correlation among numeric fields, and IQR-based outlier counts. Explain what the findings mean and distinguish correlation from causation.

6. Visualizations

The starter repository contains six example charts: monthly sales, sales by category, profit by region, top sub-categories, segment sales mix, and a correlation matrix. Rebuild all charts using the public dataset before final submission.

7. Power BI Dashboard

Build a one-page interactive dashboard with KPI cards for total sales, total profit, and distinct orders; at least four visuals; and date, region, category, and segment slicers. See powerbi/DASHBOARD_GUIDE.md for layout and DAX.

8. Business Insights and Recommendations

Replace the demo findings with at least five evidence-based insights from the public dataset. Add at least two recommendations, each linked to a specific metric or pattern.

9. Conclusion

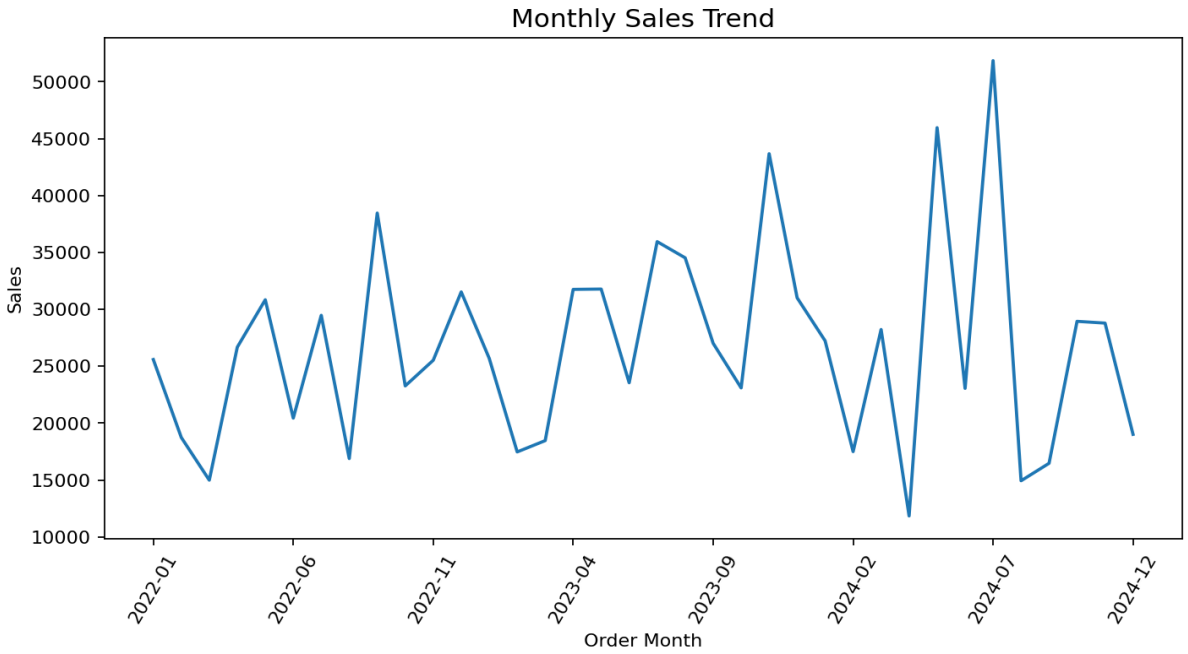
Summarize the data preparation, key patterns, business implications, and limitations. Mention that findings describe the selected educational dataset and may not generalize to every retailer.

10. Final Submission Checklist

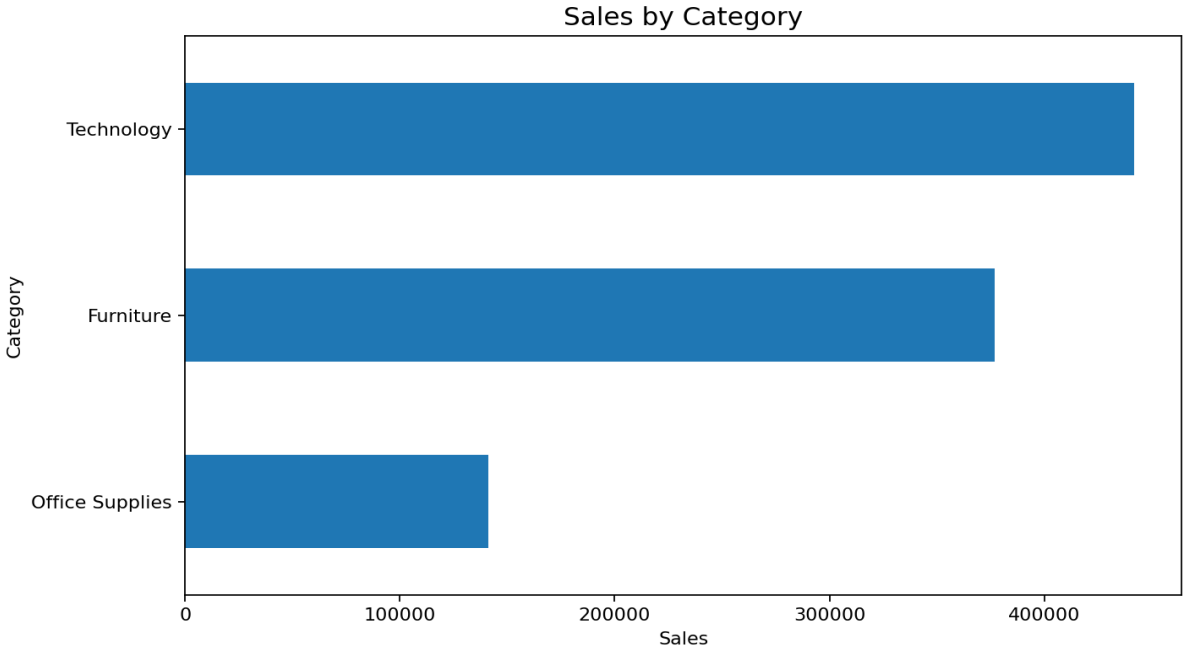
Run the notebook on the public dataset; regenerate charts; save the Power BI PBIX and screenshot; update insights; review the README; and push at least three meaningful commits.

Example charts from the simulated demo data (replace for final):

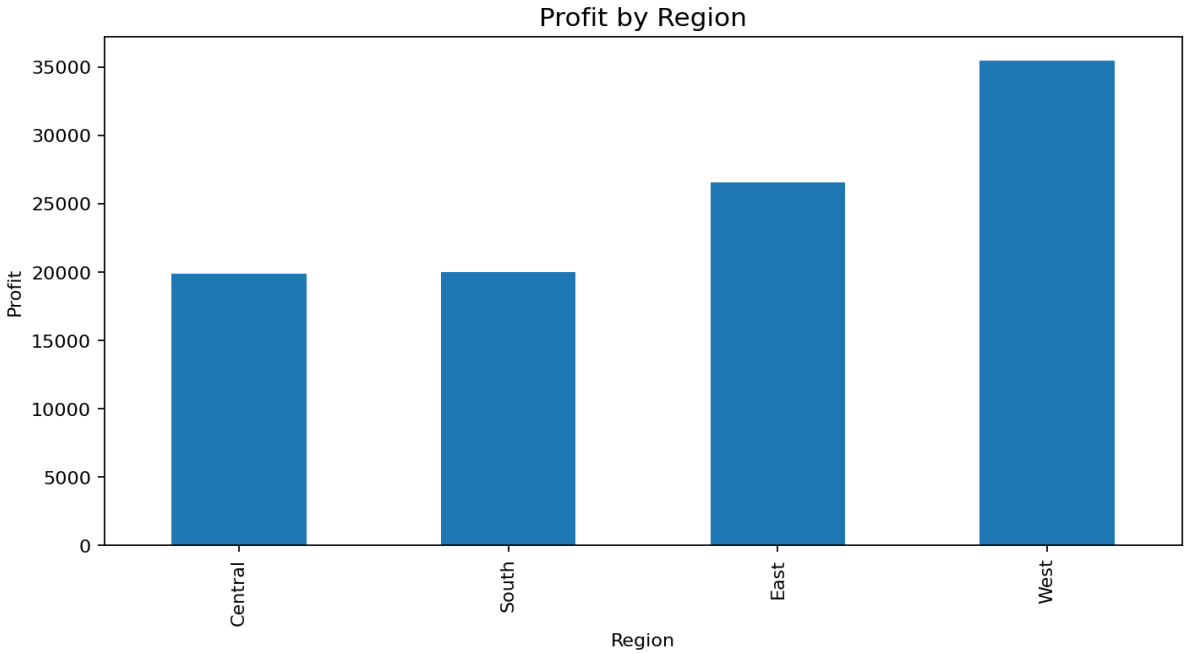
Monthly Sales



Category Sales



Profit By Region


Profit by Region
35000
30000
25000
20000
15000
10000
5000
0
Profit
Central
South
East
West
Region